Pseudocode for MULTI QUEUE Scheduling

```
#include<stdio.h>

#define MAX 20

struct Process {
    int pid, at, bt, ct, tat, wt, type;
};
int main() {
    struct Process p[MAX], sq[MAX], uq[MAX];
    int n, i, sq_count = 0, uq_count = 0, time = 0;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("\nProcess %d\n", i + 1);
        p[i].pid = i + 1;
        printf("Arrival Time: ");
        scanf("%d", &p[i].at);
        printf("Burst Time: ");
        scanf("%d", &p[i].bt);
        printf("Type (1=System,2=User): ");
        scanf("%d", &p[i].type);
        if (p[i].type == 1)
            sq[sq_count++] = p[i];
        else
            uq[uq_count++] = p[i];
    }
    for (i = 0; i < sq_count; i++) {
        if (time < sq[i].at)
            time = sq[i].at;
            time += sq[i].bt;
        sq[i].ct = time;
    }
    for (i = 0; i < uq_count; i++) {
        if (time < uq[i].at)
            time = uq[i].at;
            time += uq[i].bt;
        uq[i].ct = time;
    }
    float total_tat = 0, total_wt = 0;

    printf("\n---System Queue---\n");
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

    for (i = 0; i < sq_count; i++) {
        sq[i].tat = sq[i].ct - sq[i].at;
        sq[i].wt = sq[i].tat - sq[i].bt;
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        total_tat += sq[i].tat;
        total_wt += sq[i].wt;

        printf("%d\t%d\t%d\t%d\t%d\t%d\n",
               sq[i].pid, sq[i].at, sq[i].bt,
               sq[i].ct, sq[i].tat, sq[i].wt);
    }

    printf("\n---User Queue---\n");
    printf("PID\tAT\tBT\tCT\tTAT\tWT\n");

    for (i = 0; i < uq_count; i++) {
        uq[i].tat = uq[i].ct - uq[i].at;
        uq[i].wt = uq[i].tat - uq[i].bt;

        total_tat += uq[i].tat;
        total_wt += uq[i].wt;

        printf("%d\t%d\t%d\t%d\t%d\t%d\n",
               uq[i].pid, uq[i].at, uq[i].bt,
               uq[i].ct, uq[i].tat, uq[i].wt);
    }

    printf("\nAverage Turnaround Time=%.2f", total_tat /
    n); printf("\nAverage Waiting Time=%.2f\n", total_wt /
    n);

    printf("Naysha Navin Shrivastav\n");

    return 0;
}
```

```
Enter number of processes: 4

Process 1
Arrival Time: 0
Burst Time: 4
Type (1=System,2=User): 1

Process 2
Arrival Time: 0
Burst Time: 3
Type (1=System,2=User): 1

Process 3
Arrival Time: 0
Burst Time: 8
Type (1=System,2=User): 2

Process 4
Arrival Time: 10
Burst Time: 5
Type (1=System,2=User): 1

---System Queue---
PID     AT      BT      CT      TAT     WT
1       0       4       4       4       0
2       0       3       7       7       4
4       10      5       15      5       0

---User Queue---
PID     AT      BT      CT      TAT     WT
3       0       8       23      23      15

Average Turnaround Time=9.75
Average Waiting Time=4.75
Naysha Navin Shrivastava
```